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Community Detection in Social Networks Using a Local Approach based on Node Ranking

¹Jafar Sheykhzadeh, ²Bagher Zarei, and ¹Farhad Soleimanian Gharehchopogh

^{1,3}Department of Computer Engineering, Urmia branch, Islamic Azad University, Urmia, Iran

²Faculty of Computer Engineering and Information Technology, Shabestar branch, Islamic Azad University, Shabestar, Iran

Corresponding author: Bagher Zarei (e-mail: zareibager@iau.ac.ir).

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ABSTRACT Community detection is crucial for analyzing the structure of social networks and extracting hidden information from them. The goal is to find groups of nodes (communities) with high intra-group and low inter-group communications. This problem is NP-hard, and most existing algorithms are global with high computational complexity, especially for large networks. Recently, local methods with acceptable computational complexity have been developed, but many have low accuracy and are non-deterministic. This paper introduces a new local algorithm, LCD-SN, which identifies communities based on first- and second-degree neighbor nodes. Unlike other local algorithms, LCD-SN is highly accurate, definitive, and not dependent on initial seed nodes. Additionally, a new index is proposed to determine the importance of network nodes, using their local characteristics (first- and second-degree neighbors). Using this index, LCD-SN first identifies important nodes, forms initial communities with these nodes and their first-degree neighbors, and then obtains final communities through post-processing. Experiments show that LCD-SN is effective in identifying communities in social networks.

INDEX TERMS Community Structure Detection, Modularity, Nodes Ranking, Social Network Analysis

I. INTRODUCTION

Many existing systems can be modeled using graphs. In mathematics, a graph is shown as $G = (V, E)$, where $V = \{v_1, v_2, \dots, v_n\}$ is a set of $|V| = n$ vertices (nodes) and $E \subseteq V \times V$ is a set of $|E| = m$ edges. For example, in a social network graph, nodes show people and edges show connections between them. All nodes connected to the vertex v_i are called its neighborhood set and are denoted by $\Gamma(v_i)$ [1].

The identification of communities is one of the main problems in the field of social network analysis. Identifying communities makes it possible to extract hidden information in the network and obtain information about people or entities. The community structure detection problem aims to identify groups of nodes so that there are relatively more connections within each group and relatively fewer connections between the groups [2, 3]. So far, many criteria have been proposed to evaluate the structure of the community. One of the most well-known evaluation criteria is the modularity criterion presented by Girvan Newman [4].

The community structure detection problem can be defined as a modularity optimization problem. It has been proven that modularity optimization is an NP-hard problem [2].

The existing algorithms for discovering the community structure are divided into global and local categories. Global algorithms are more accurate, but due to the need to access complete network information, their time complexity is high and cannot be used in large networks. In contrast, local algorithms have less time complexity due to limited information [5, 6]. However, local algorithms have disadvantages, such as their dependence on predefined parameters, instability, and low quality.

In recent years, the use of clustering algorithms to detect communities in social networks has attracted the attention of researchers. Due to the lack of access to complete information on social networks, local methods are used to increase efficiency and speed in large-scale networks. In most local methods, communities are determined by selecting important nodes as the core of communities. Most of the existing works have disadvantages such as low

accuracy due to inappropriate core selection, lack of scalability, and uncertainty in the results. In the proposed algorithm, a new index is proposed to calculate the importance of network nodes based on the structural characteristics of the network. The proposed index reflects the real position of the nodes by considering the first- and second-degree neighbors. After determining the importance of nodes, important nodes are considered as the core of communities. Initial communities are formed around the core nodes based on the structural characteristics of the network. Finally, with a post-processing the final communities are obtained. In addition to having the advantages of existing local methods, the proposed algorithm solves the issues of local methods, including the dependence of communities on core nodes, uncertainty, and low quality. The results of experiments show that the proposed algorithm performs better than the compared algorithms in most cases. The main contribution of this article is as follows:

- A new ranking index called IMP is presented. The IMP index is a local index that determines the importance of nodes based on the first- and second-degree neighborhood.
- A new local algorithm called LCD-SN is presented, which can identify communities with high accuracy using the IMP index.
- The generalized Leicht-Holme-Newman similarity index assigns overlapping nodes to a single community.
- The results of experiments in synthetic and real-world benchmark networks show that the introduced algorithm performs better than other algorithms.

The remainder of this article is organized as follows: Section 2 reviews some related works. The introduced method is provided in Section 3. Section 4 includes experimental results and compares the presented method with some known methods. Finally, Section 5 consists of the conclusion and summary of the article.

II. RELATED WORKS

In a general classification, existing community structure detection algorithms are divided into global and local categories. Global algorithms are more accurate, but due to the need to access complete network information, their time complexity is high and cannot be used in large networks. In contrast, local algorithms have less time complexity due to the use of limited information. However, local algorithms have disadvantages, such as their dependence on predefined parameters, instability, and low quality. The following section has reviewed some of the most well-known global and local algorithms.

A. GLOBAL COMMUNITY STRUCTURE DETECTION ALGORITHMS

The methods based on graph partitioning are among the first global methods for detecting the structure of communities. These methods divide the graph of a social network into k predefined parts (communities) so that the sum of edges between communities is minimized. Graph partitioning methods are inappropriate for discovering communities in large networks due to high computational complexity. The Kernighan-Lin method is one of the most famous algorithms for this category [7].

Hierarchical clustering methods are divided into agglomerative and divisive categories, among other global community detection methods. Divisive hierarchical clustering methods cluster the graph by identifying and removing inter-cluster edges. The algorithm of Girvan-Newman [4, 8] is one of the most popular divisive hierarchical algorithms. This method identifies and removes inter-cluster edges using the edge betweenness centrality. An edge with the highest amount of betweenness is a bridge between communities. This algorithm's time complexity is $O(n^3)$. In [9], communities have been identified using the agglomerative hierarchical method based on modularity criterion and cosine similarity index. These methods are only applicable in small networks due to high time complexity.

In [10], a modularity-based heuristic method called Louvain has been introduced. In this method, each node is initially considered an independent community. Then, during an iterative process, those nodes whose merging increases the modularity criterion are merged. The merging of nodes continues until no improvement in modularity is achieved. After forming initial communities, a new graph is created in which each node corresponds to a community of initial communities, and the edge weight between two nodes is equal to the number/sum of the weights of the edges between the two communities corresponding to them. The above steps (merging and forming a new graph) are repeated on the new graph. Merging and creating a new graph continues until no improvement in modularity is achieved.

In [11] and [12], a genetic algorithm named GACD and GATB has been proposed for community discovery. These algorithms are according to modularity optimization. In these algorithms, locus-based adjacency coding is used to represent a graph partition. Locus-based adjacency coding for community detection has the following benefits: (1) its search space is smaller than string coding, (2) the community number is automatically specified in the decoding process, and (3) the decoding process is very efficient. Crossover and mutation operators do not lead to invalid solutions in this coding.

In [13], three algorithms named MEM-net, OMA-net, and GAOMA-net have been introduced to detect community structures in complex networks. These algorithms do not require previous information, such as the community

number, and identify communities dynamically. In the GAOMA-net algorithm, which is the main algorithm introduced in this paper, the Object Migration Automata (OMA) and genetic algorithm are combined as a single framework, and the algorithm of MEM-net is utilized as a heuristic to generate a part of the initial population. This combination accelerates the algorithm convergence and prevents it from falling into the local optimum.

In [14], a chaotic memetic algorithm (CMA) has been introduced to identify communities. In the CMA algorithm, a combination of dedicated local search and genetic algorithm (global search) has been utilized to search the solution space. In addition, this article uses chaotic numbers instead of random numbers. Chaotic numbers preserve the diversity of the population and avoid the algorithm falling into the local optimum.

In [15], a chaotic cellular learning automata-based evolutionary model (CCLA-EM) was presented to discover communities in complex networks. CCLA-EM algorithm combines an evolutionary algorithm with cellular learning automata. In this algorithm, the individuals of the population are distributed over the cells of a cellular learning automata. Each individual cooperates and communicates with the individuals of the neighboring cells to achieve the global optimum. Individual distribution over cells in cellular learning automata causes the parallel implementation of the presented model. In addition, this method uses chaotic numbers instead of random numbers. Using chaotic numbers instead of random numbers causes a complete search in the search space and prohibits the algorithm from getting trapped in the local optimum.

B. LOCAL COMMUNITY STRUCTURE DETECTION ALGORITHMS

Label Propagation Algorithm (LPA) [16] is one of the most famous local community discovery algorithms proposed by Raghavan. In this algorithm, first, each node is given a unique label. Then, each node label is updated by majority voting on the labels of its neighbors. The node labels update process continues until no change occurs in the node labels. The most significant advantage of LPA is its linear computational complexity, and one of its disadvantages is its uncertainty. In [17], the problem of indeterminacy and randomness of the LPA has been overcome using the propagation of the label from the least important node.

In [18], using label propagation and MinHash, an algorithm called WLPA has been presented for community detection in unsigned and signed networks. In the WLPA algorithm, communication intensity is also considered in addition to communication. In this method, the similarity of all adjacent nodes is determined using MinHash. Therefore, each edge is given a weight equal to the similarity of its end nodes. The weight given to each edge indicates the intensity of the connection between its end nodes. Finally, the community structure is specified through the propagation of

weighted labels. The results of the experiments show that using the WLPA algorithm is effective and efficient for discovering communities in signed and unsigned networks.

In [19], a four-step method has been presented for extracting communities. In the first step, important nodes are found by extracting global and local structures and using game theory. In the second stage, with the propagation of the label, primary communities are formed. In the third stage, the obtained communities are integrated. Finally, in the fourth step, the final communities are extracted by ensuring the correct allocation of nodes. This method has low accuracy in detecting communities in real-world networks.

In [20], a three-step method has been presented for local community detection in social networks. These three steps include (1) determining the importance of nodes and forming primary communities, (2) extending primary communities, and (3) merging primary communities to obtain final communities. In the first stage, the importance of network nodes is calculated using an index called RDC. Then, initial communities are formed, including several important nodes. In the second stage, the initial communities are extended using a local similarity measure so that each network node becomes a community member. The extracted communities are investigated in the third stage, and small communities are merged with other communities.

In [21], a two-stage local community discovery algorithm, RTLCD, based on community expansion and core detection, has been presented. Local community discovery algorithms have two common challenges and issues: (1) seed-dependent problem, meaning whether the seed node location affects the quality of the detected local community or not, and (2) the invalid core problem, which means that some local algorithms cannot ensure the core node participates in the final community. This article used the criteria of node relation strength and node mass (node similarity and local indices for node centrality), as well as community relationship strength, to solve the two mentioned problems.

In [22], a three-stage local community detection algorithm called Three-Stage (TS) has been presented based on global and local information. These steps include central node recognition, label propagation, and community integration. Central nodes are determined based on the distance between them (greater than the average distance). Label propagation means labeling nodes with identical colors when they reach maximum similarity. The integration of communities is done if the modularity increase is positive and at its maximum level.

In [23], a local three-stage algorithm called LCDPC has been presented for community detection, which identifies local communities based on exploring potential communities. In this algorithm, first, an appropriate node is found to replace a particular node as a seed by determining the similarity and importance of the node. Then, the initial community is formed by the composition of the seed and its

appropriate potential community. Eventually, the eligible nodes are chosen by comparing the similarity between expanding nodes and potential communities to add to the initial community. This algorithm has a good speed in detecting communities, but the quality of the extracted communities is similar to other algorithms.

In [24], a novel approach to community discovery is proposed by considering the importance score and membership degree of each node. Nodes with higher importance scores can become the core of communities, leading to the formation of communities with a reasonable number of nodes.

In [25], has been introduced a group clustering algorithm based on influential nodes to improve the detection of communities in complex networks. Considering the diverse attributes of the network, the proposed approach searches to find shared interests among users and their behaviors to identify the most appropriate communities. Initially, a group of influential nodes are recognized as community centers. Then, initial communities are created based on the determined centers. Finally, the initial communities are reclustered to form the final communities.

In [26], a novel approach called Deep Semi-Supervised Community Detection (DSSC) has been presented for complex network clustering. DSSC utilizes a semi-autoencoder (SeAE) with a pair-wise constraint matrix based on point-wise mutual information (PMI) to learn distinctive features and enhance clustering accuracy.

In [27], by combining deep learning and random walk, a new method called modified DeepWalk has been presented. It employs a modified Random Walk with Restart to capture k-order structural and attribute information, enabling the modeling of interactions between network structure and high-order proximities.

In [28], an extension of Symmetric Nonnegative Matrix Factorization (SNMF) called Weighted Symmetric Nonnegative Matrix Factorization (WSNMF) is designed to address challenges encountered in attributed graph clustering. While SNMF has shown promise in graph clustering, it lacks consideration for attributed information, geometric data point structures, and the ability to discriminate irrelevant features and outliers.

In [29], a rapid and precise community detection method in large networks based on locally balanced label propagation is proposed. A novel local similarity measure is also presented to assign importance to nodes. This method demonstrates fast convergence speed along with stable and accurate results.

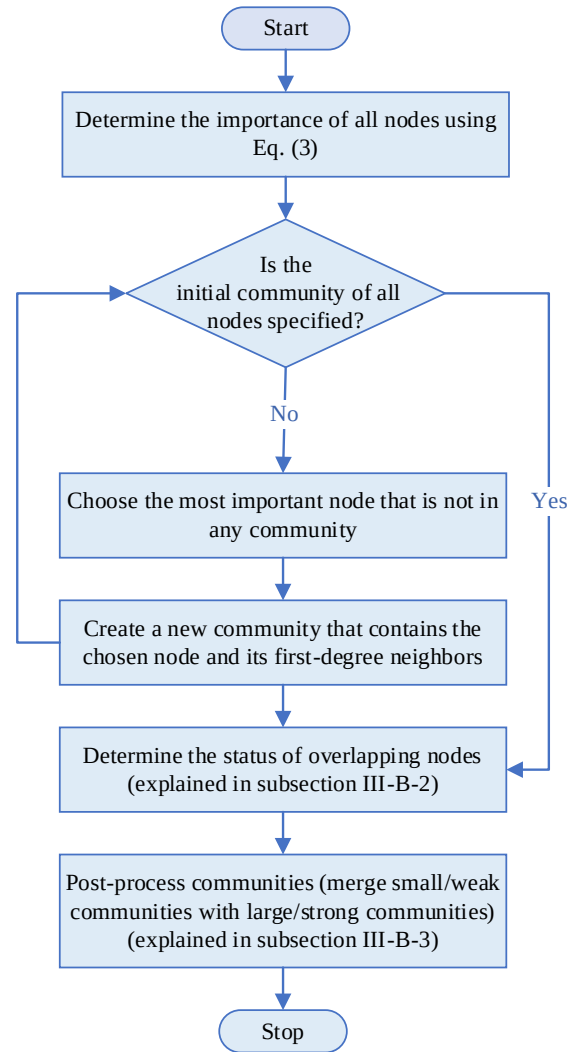


FIGURE 1. Flowchat of the LCD-SN algorithm

TABLE I. Notations used in the proposed algorithm

Notation	Description
$IMP(i)$	Importance of node i
r_i^{in}	The set of nodes linked to i
w_{ji}	The directed edge weight from node j to node i
w_j^{out}	The sum of the weight of outgoing edges from node j
α	Influence of the first-degree neighbors on the importance of a node
β	Influence of the second-degree neighbors on the importance of a node
γ	The maximum number of iterations for calculating the importance of nodes
$N_i^{1,2}$	The set of first- and second-degree neighbors of node i
C_i	i^{th} community
E_i^{in}	Number of edges inside i^{th} community
E_i^{out}	Number of edges from i^{th} community to other communities
m_c	The integration coefficient of the communities
A_{ij}	The element of i^{th} row and j^{th} column of the adjacency matrix A
P_{ij}	The expected value of the number of edges between two nodes i and j

III. THE PROPOSED ALGORITHM

In this section, a new index called IMP has been presented to determine the importance of nodes. Then, using the IMP index, a local algorithm called LCD-SN has been presented to determine communities in social networks. The flowchart of the proposed method is illustrated in Figure 1. In addition, the notations used in presenting the proposed algorithm are listed in Table I.

A. IMP IMPORTANCE INDEX

This subsection presents a new index called IMP to determine the importance of nodes according to the nodes' first- and second-degree neighbors. In the IMP index, structural and local characteristics of nodes have been used to determine their importance. The law of continuity of electric current in electric circuits inspires this index. According to the continuity law of electric current, the totality of currents entering a node equals the totality of currents leaving that node. In other words, for each node of an electric circuit, we have:

$$\sum I^{in} = \sum I^{out} \quad (1)$$

The graph of a directed and weighted social network can be considered as corresponding to an electrical circuit where (a) each of its nodes corresponds to a node of the electrical circuit and (b) the importance of a node is equal to the sum of the weighted incoming/outgoing electric currents to/from that node. Considering that in real-world networks, the importance of a node is affected by the nodes that have a link to it and the intensity of the link (connection weight) between the nodes, Eq. (2) can be used to determine the importance of a node:

$$IMP(i) = \sum_{j \in \Gamma_i^{in}} \frac{W_{ji} \times IMP(j)}{W_j^{out}} \quad (2)$$

In Eq. (2), Γ_i^{in} is the set of nodes linked to i , W_{ji} is the directed edge weight from node j to node i , and W_j^{out} is the sum of the outgoing edge weight from node j . For example, consider Figure 2, which represents a portion of a social network graph. Suppose the importance of nodes x , y , and z equals 10, 20, and 30, respectively. According to Eq. (2), the importance of node i is equal to:

$$IMP(i) = \frac{2 \times 10}{5} + \frac{1 \times 20}{2} + \frac{2 \times 30}{6} = 24$$

Considering that we initially do not know the importance of nodes, the initial importance of all nodes is assumed to be the same. There may also be nodes whose importance is mutually dependent (nodes i and x in Figure 2). Therefore, calculating the node's importance is repeated until either the importance of the nodes converges or the process of calculating the node's importance is repeated a maximum of γ times. The Eq. (2) is defined only according to the first-degree neighbors of the nodes. To increase the accuracy, in

addition to the first-degree neighbors, second-degree neighbors can also be considered. In this case, Eq. (2) will be generalized as

$$IMP(i) = \alpha \sum_{j \in \Gamma_i^{in}} \frac{W_{ji} \times IMP(j)}{W_j^{out}} + \beta \sum_{k \in \Gamma_j^{in}} \frac{W_{kj} \times IMP(k)}{W_k^{out}} \quad (3)$$

In Eq. (3), α and β indicate the influence of first- and second-degree neighbors on the importance of node i .

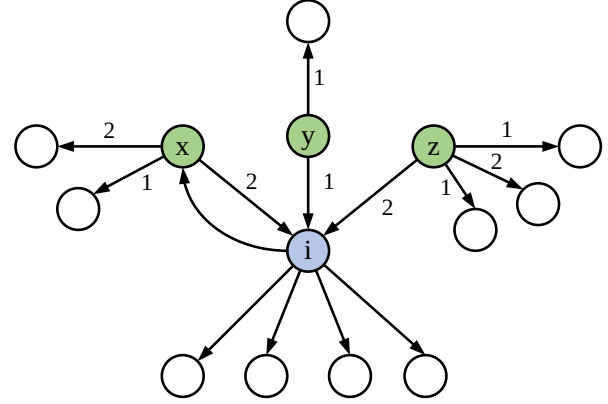


FIGURE 2. A part of the graph of a directed and weighted social network. According to Eq. (2), the importance of node i depends on the importance of nodes x , y , and z .

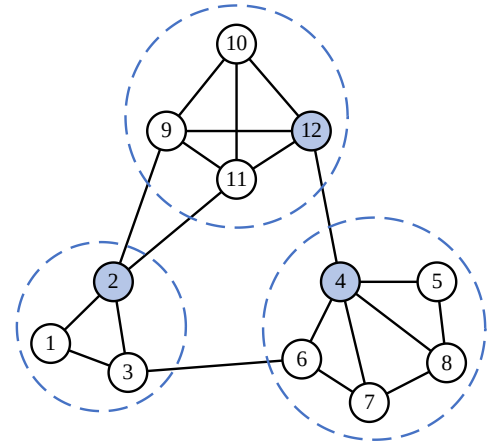


FIGURE 3. An example network having 12 nodes and three communities

TABLE II. Ranking of network nodes in Figure 3 using IMP, PageRank, and degree criteria

IMP		PageRank		degree	
Rank	Value	Rank	Value	Rank	Value
4	2.5250	4	0.1227	4	5
12	2.1233	2	0.0985	2	4
2	2.0833	9	0.0936	9	4
9	2.0183	11	0.0936	11	4
11	2.0183	12	0.0935	12	4
6	1.7167	3	0.0793	3	3
7	1.7017	8	0.0792	6	3
8	1.6683	6	0.0779	7	3
3	1.5933	7	0.0779	8	3
10	1.4850	10	0.0721	10	3
5	1.2083	1	0.0559	1	2
1	1.1333	5	0.0558	5	2

In Table II, using IMP, PageRank [30], and Degree [31] indices, the importance of the Figure 3 network nodes has been presented. As observed, nodes 4, 12, 2, 9, and 11 were the five nodes with the highest rank in all three indices. This shows that the IMP index aligns with other important indices in the literature, such as PageRank and Degree index. Nodes 4, 12, and 2 (the three nodes with the highest rank using the IMP index) could be used as seeds in the proposed algorithm to form initial communities.

B. LCD-SN ALGORITHM FOR COMMUNITY STRUCTURE DETECTION

In this subsection, an algorithm called LCD-SN has been presented to discover community structure in social networks. The LCD-SN algorithm consists of three phases: (1) forming initial communities, (2) determining the status of overlapping nodes, and (3) merging communities. In the following, each of these phases has been explained in detail.

1) THE FIRST PHASE: FORMATION OF INITIAL COMMUNITIES

In this phase, the network nodes are first ranked using the Eq. (3). Then, initial communities are formed using the following procedure:

- (1) The node with the highest rank value, which is not already in any community, is selected as the community's core. Suppose the selected node is k .
- (2) Node k , along with its first-degree neighbors, is considered a community.
- (3) Steps 1 and 2 are repeated until the initial community of all nodes is determined.

In this phase, some nodes may be members of several communities. In other words, initial communities may overlap. By applying this phase on the Karate network (Figure 4(a)), four initial communities $C1=\{1, 2, 3, 4, 5, 6, 7, 8, 9, 11, 12, 13, 14, 18, 20, 22, 32\}$, $C2=\{34, 9, 10, 14, 15, 16, 19, 20, 21, 23, 24, 27, 28, 29, 30, 31, 32, 33\}$, $C3=\{26, 24, 25, 32\}$, and $C4=\{17, 6, 7\}$ are obtained. These communities are shown in various colors in Figure 4(b). Note that nodes 6, 7, 9, 14, 20, 24, and 32 overlap and are shown in black.

2) SECOND PHASE: DETERMINING THE STATUS OF OVERLAPPING NODES

In this phase, overlapping nodes are placed only in one of the communities and are removed from the rest of the communities. For this purpose, the similarity of each overlapping node with all the communities to which it belongs is calculated. The overlapping node is placed within the community with the most similarity. In this article, the Generalized Leicht-Holme-Newman (GLHN) similarity index [32] is used to calculate the similarity of a node to a community. In GLHN, to identify the similarity of two nodes, in addition to the first-degree neighborhood, their

second-degree neighborhood is also considered. This index has been given in Eq. (4).

$$GLHN(i, j) = \frac{|N_i^{1,2} \cap N_j^{1,2}|}{|N_i^{1,2}| * |N_j^{1,2}|} \quad (4)$$

In Eq. (4), $N_i^{1,2}$ represents the set of first- and second-degree neighbors of node i . Using Eq. (4), the similarity of node i to community C is defined as Eq. (5).

$$sim(i, C) = \sum_{\substack{j \in C, \\ j \in \Gamma_i}} GLHN(i, j) \quad (5)$$

By applying this phase to the output of the first phase (Figure 4(b)), the state of the nodes and communities of Zachary's karate club will be as shown in Figure 4(c). As observed, overlapping nodes 6, 7, 14, and 20 are assigned to community $C1$, and the rest (nodes 9, 24, and 32) are assigned to community $C2$.

3) THE THIRD PHASE: INTEGRATION OF COMMUNITIES

Considering that some of the communities obtained in the previous phase may be too small or weak, in this phase, the communities obtained in the previous phase are merged in two stages so that the final communities are obtained: (1) the integration of small communities with large communities, and (2) the integration of weak communities with strong communities. In merging small communities with large communities, first small communities (communities with less than 3 nodes) are identified and then each node of them is transferred to one of the large neighboring community that is most similar to it. It should be noted that Eq. (5) is used to calculate the similarity of a node to a community.

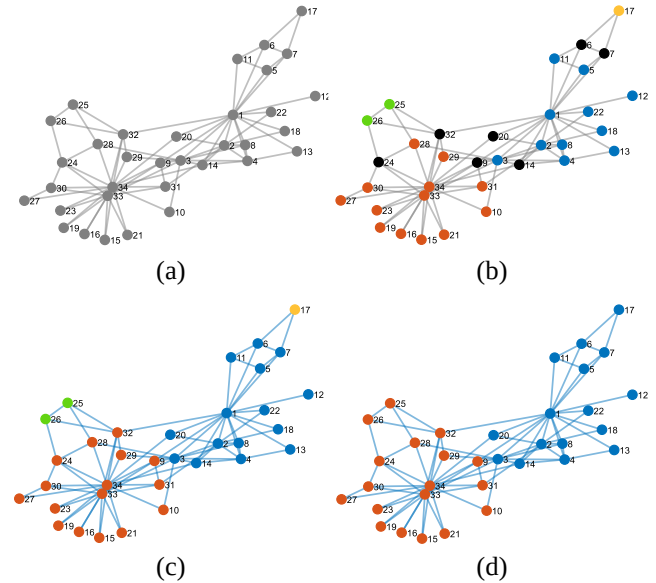


FIGURE 4. (a) Zachary's karate club network and (b)/(c)/(d) the result of the first/second/third phase of the proposed algorithm on it

Algorithm 1: LCD-SN (Local Community Detection in Social Networks)

Input: G : Social Network Graph
 α : Influence of first-degree neighbors on the importance of a node
 β : Influence of second-degree neighbors on the importance of a node
 γ : Maximum number of iterations for calculating the importance of a node

Output: Community structure of the input graph G

// First Phase-Formation of Initial Communities
1 Calculate the importance of all nodes of the input graph G using Eq. 3;
2 **repeat**
3 Choose the most important node that is not in any community;
4 Create a new community that contains the chosen node and its first-degree neighbors;
5 **until** the initial community of all the nodes is determined;
// Second Phase-Determining Status of Overlapping Nodes
6 **foreach** overlapping node v **do**
7 Calculate the similarity of node v with all the communities it belongs to using Eq. 5;
8 Place node v in the community that is most similar to it;
// Third Phase-Integration of Communities
// Merge small communities with large ones
9 $L \leftarrow$ nodes of small communities (communities with less than 3 nodes);
10 **repeat**
11 $v \leftarrow$ Select a node from list L ;
12 Merge node v with the neighboring community that is most similar to it (Calculate the similarity of a node to a community using Eq. 5);
13 **until** list L is empty;
// Merge weak communities with strong ones
14 **foreach** community C_i **do**
15 **if** $E_i^{in} \leq m_c \times E_i^{out}$ (C_i is a weak community) **then**
16 Move the nodes of community C_i to the neighboring community that is most similar to it (Calculate the similarity of two communities using Eq. 6);

TABLE III. Parameters used in different algorithms

Algorithm	Parameter	Value
CMA [14]	Population size	100
	Crossover rate	0.8
	Mutation Rate	0.2
	Elitism rate	0.05
	Max generations	100
	Local search rate	0.1
	Logistic map parameter	4
GAOMA-net [33]	Population size	100
	Recombination rate	0.8
	Mutation rate	0.2
	Elitism rate	0.05
	Max generations	100
	Object migrating automaton type	Tsetlin
	Actions memory depth	5
GACD [11]	Population size	100
	Crossover rate	0.8
	Mutation rate	0.2
	Max generations	100
	Population size	100
GATB [12]	Crossover rate	0.8
	Mutation rate	0.2
	Initial rate	0.6
	Elitism rate	0.1
	Clean rate	0.5
	Clean threshold	0.7
	Max generations	100
	Inflation power	4
LabelRank [34]	Cutoff threshold	0.1
	Conditional update coefficient	0.7
	Max Iterations	50
LBLD [29]	-	-
CSLPR [17]	-	-
LCD-SN	α	0.7
	β	0.3
	γ	6

Some extracted communities may not be of good quality. To increase the quality of communities, weak communities should be integrated with strong communities. In this paper, a group of nodes is considered a weak community if the condition $E_i^{in} \leq m_c \times E_i^{out}$ is satisfied. In this equation, E_i^{in} , E_i^{out} , and m_c , respectively, indicate the edge number inside the community, the edge number outside the community, and the integration coefficient of the communities. The process of merging weak communities with strong communities continues until there are no weak communities. The nodes of a weak community are transferred to strong neighboring community that are most similar to it. It should be noted that two communities are said to be neighbors if there is at least one edge between them. We can use Eq. (6), a generalization of Eq. (5), to calculate the similarity of the two communities. It should be noted that integration is carried out if it leads to increased modularity. Based on the experimental results, by choosing $m_c = 4$, the final community structure will have a favorable quality.

$$sim(C_1, C_2) = \sum_{\substack{i \in C_1, \\ j \in C_2, j \in \Gamma_i}} GLHN(i, j) \quad (6)$$

By applying this phase to the output of the second phase of the proposed algorithm (Figure 4(c)), the only remaining member of community C_4 (node 17) is merged with community C_1 , and the remaining nodes of community C_3 (nodes 25 and 26) are merged with community C_2 . It should be noted that two large communities will be formed after the

integration of small communities (Figure 4(d)), none of which could meet the condition of a weak community, so the third phase is completed. The pseudocode of the proposed algorithm is given in Algorithm 1.

IV. EXPERIMENTAL RESULTS AND DISCUSSION

In this section, the efficiency of the introduced algorithm is evaluated on real-world and synthetic benchmark networks and is compared with CMA [14], GAOMA-net [33], GACD [11], GATB [12], LabelRank [34], LBLD [29], and CSLPR [17] algorithms.

A. PARAMETER SETTING

The proposed and compared algorithms are implemented using MATLAB R2022a and run on a personal computer with a core i9-13900K 3.00 GHz processor and 64GB of RAM. Table III gives the values of parameters used in different algorithms.

B. EVALUATION CRITERIA

To evaluate and compare the algorithms, two measures of Q (modularity) [4] and NMI (Normalized Mutual Information)

TABLE IV. The results obtained from the proposed algorithm for Q criterion in real-world benchmark networks for $\alpha=0.7$, $\beta=0.3$, and different values of γ parameter

Network	$\gamma = 1$	$\gamma = 2$	$\gamma = 3$	$\gamma = 4$	$\gamma = 5$	$\gamma = 6$	$\gamma = 7$	$\gamma = 8$	$\gamma = 9$	$\gamma = 10$
Karete	0.3715	0.3715	0.3715	0.3715	0.3715	0.3715	0.3715	0.3715	0.3715	0.3715
Dolphins	0.5087	0.5187	0.5005	0.5005	0.5005	0.5005	0.5005	0.5005	0.5005	0.5005
PolBooks	0.5240	0.5240	0.5240	0.5240	0.5240	0.5240	0.5240	0.5240	0.5240	0.5240
Football	0.5962	0.5815	0.6056	0.6056	0.6056	0.6056	0.6056	0.6056	0.6056	0.6056
SFI	0.6918	0.6918	0.6918	0.6918	0.6918	0.6918	0.6918	0.6918	0.6918	0.6918
NetScience	0.8434	0.8432	0.8435	0.8435	0.8435	0.8435	0.8375	0.8375	0.8375	0.8375
Email	0.5257	0.5297	0.5279	0.5279	0.5266	0.5284	0.5258	0.5354	0.5354	0.5354
PowerGrid	0.9146	0.9101	0.9123	0.9140	0.9137	0.9159	0.9122	0.9124	0.9114	0.9107
PGP	0.8584	0.8578	0.8588	0.8626	0.8625	0.8629	0.8624	0.8619	0.8621	0.8620
GrQc	0.8270	0.8244	0.8252	0.8246	0.8257	0.8277	0.8252	0.8268	0.8258	0.8250
ca-AstroPh	0.5567	0.5574	0.549	0.5542	0.5527	0.5629	0.5563	0.5560	0.5557	0.5557
ca-HepTh	0.7255	0.7289	0.7330	0.7311	0.7287	0.7331	0.7323	0.7269	0.7340	0.7333
ca-HepPh	0.6074	0.6124	0.6052	0.6085	0.6090	0.6138	0.6129	0.6130	0.6130	0.6130
Condmatt-2003	0.6821	0.6872	0.6807	0.6809	0.6857	0.6776	0.6758	0.6751	0.6785	0.6780
Condmatt-2005	0.6425	0.6315	0.6385	0.6441	0.6382	0.6311	0.6276	0.6236	0.6176	0.6277
Email Enron	0.5349	0.5466	0.5352	0.5390	0.5299	0.5244	0.5132	0.5351	0.5357	0.5353
Collaboration	0.7914	0.7944	0.7995	0.8012	0.8027	0.7967	0.7972	0.8011	0.7982	0.7948
Internet	0.5497	0.5315	0.5010	0.5004	0.4983	0.4981	0.4981	0.4981	0.4981	0.4981

TABLE V. The results obtained from the introduced algorithm for the NMI criterion in LFR synthetic benchmark networks for $\alpha = 0.7$, $\beta = 0.3$, and different values of the γ parameter

μ	$\gamma = 1$	$\gamma = 2$	$\gamma = 3$	$\gamma = 4$	$\gamma = 5$	$\gamma = 6$	$\gamma = 7$	$\gamma = 8$	$\gamma = 9$	$\gamma = 10$
0.00	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
0.05	0.9902	0.9920	0.9914	0.9909	0.9914	0.9922	0.9915	0.9915	0.9922	0.9928
0.10	0.9831	0.9814	0.9794	0.9788	0.9798	0.9805	0.9800	0.9800	0.9783	0.9783
0.15	0.9652	0.9698	0.9691	0.9697	0.9738	0.9744	0.9744	0.9744	0.9729	0.9729
0.20	0.9520	0.9535	0.9519	0.9552	0.9561	0.9569	0.9569	0.9569	0.9569	0.9569
0.25	0.9278	0.9278	0.9258	0.9257	0.9257	0.9257	0.9257	0.9257	0.9257	0.9257
0.30	0.9044	0.9067	0.9067	0.9078	0.9078	0.9078	0.9078	0.9078	0.9078	0.9078
0.35	0.8352	0.8473	0.8462	0.8485	0.8487	0.8487	0.8487	0.8487	0.8487	0.8487
0.40	0.7494	0.7388	0.7340	0.7346	0.7345	0.7345	0.7345	0.7345	0.7345	0.7345
0.45	0.6147	0.6098	0.6168	0.6196	0.6196	0.6196	0.6196	0.6196	0.6196	0.6196
0.50	0.4678	0.4556	0.4725	0.4566	0.4538	0.4533	0.4533	0.4533	0.4533	0.4533

$$Q = \frac{1}{2m} \sum_{ij} (A_{ij} - P_{ij}) \delta(C_i, C_j) \quad (7)$$

where m is the number of edges, A is the adjacency matrix, $P_{ij} = \frac{k_i k_j}{2m}$ is the expected value of the number of edges between two nodes i and j , C_i is the community of node i , and the function δ is defined as Eq. (8).

$$\delta(x, y) = \begin{cases} 1 & x = y \\ 0 & \text{otherwise} \end{cases} \quad (8)$$

The modularity criterion is in the range of $[-0.5, 1]$, and larger values show that the edge density within communities is higher than the null model.

The NMI criterion is based on the information theory, which compares the community's quality in terms of its compatibility with the ground-truth community. Assuming to have the found community structure $A = \{C_1, C_2, \dots, C_A\}$, and the ground-truth community structure $B = \{C'_1, C'_2, \dots, C'_B\}$, the NMI criterion then will be defined as Eq. (9).

$$\text{NMI}(A, B) = \frac{-2 \sum_{i=1}^{|A|} \sum_{j=1}^{|B|} n_{ij} \log \left(\frac{n_{ij} n}{n_i n_j} \right)}{\sum_{i=1}^{|A|} n_i \log \left(\frac{n_i}{n} \right) + \sum_{j=1}^{|B|} n_j \log \left(\frac{n_j}{n} \right)} \quad (9)$$

The NMI measure is based on the confusion matrix. The confusion matrix rows are related to the ground-truth communities, and its columns are related to the found communities. The element n_{ij} of the confusion matrix shows the number of common nodes in the ground-truth community i and the found community j , and $n_i(n_j)$ represents the sum of the i row (j column) elements of the confusion matrix. The NMI criterion is in the range $[0, 1]$. Larger values show that the found community structure is more consistent with the ground-truth community structure.

C. DETERMINING THE OPTIMAL VALUE α , β , AND γ PARAMETERS

In this subsection, to determine the optimal value of α , β , and γ parameters, the proposed algorithm with different combinations $\alpha = \{0.1, 0.2, \dots, 1\}$, $\beta = 1 - \alpha$ and $\gamma = \{1, 2, \dots, 10\}$ in Eq. (3) has been applied to various real-world and synthetic benchmark networks. The results of the experiments show that the introduced algorithm performs better for values of $\alpha = 0.7$, $\beta = 0.3$, and $\gamma = 6$. Therefore, in the results stated in the rest of this article, the values of 0.7, 0.3, and 6 have been considered for α , β , and γ parameters, respectively. For example, in Tables IV and V, the results of the introduced algorithm for $\alpha = 0.7$, $\beta = 0.3$, and various values of the γ parameter are given in real-world and synthetic benchmark networks, respectively. As you observe, the introduced algorithm performs better when $\gamma = 6$.

D. EVALUATION WITH REAL-WORLD BENCHMARK NETWORKS

In this section, the efficiency of the introduced algorithm is evaluated in real-world benchmark networks and is compared with CMA [14], GAOMA-net [13], GACD [11], GATB [12], LabelRank [34], LBLD [29], and CSLPR [17] algorithms. The information on the real-world benchmark networks is given in Table VI. The Q criterion is used to evaluate and compare different algorithms because the ground-truth community structure is not attainable in most real-world networks. The average modularity obtained from different algorithms is given in Table VII. In the non-deterministic algorithms (CMA, GAOMA-net, GACD, and GATB), each data is the average of 10 runs. The largest modularity value obtained for each network is written in bold. The LCD-SN has obtained the highest modularity value in PolBooks, Football, PowerGrid, PGP, GrQc, AstroPh, ca-HepTh, ca-HepPh, and Collaboration networks. CMA in the Dolphins and Internet networks, GAOMA-net in the Karate network, GACD in the Karate and SFI networks, LBLD in the NetScience, Email, Condmat 2003, and Condmat 2005 networks, CSLPR in the Email Enron network have obtained the highest modularity value.

In addition, Table VIII shows the results of the Friedman test for ranking different algorithms in the real-world networks. As you can see, the Friedman test statistic value (p -value=0.001) is less than 0.05. Therefore, with a probability of 0.95, it could be mentioned that the average ranks in real-world networks had statistically significant differences. According to the average ranks, the LCD-SN algorithm won first, and the LBLD and CMA algorithms won second and third, respectively. In summary, it can be said that, the LCD-SN algorithm is effective for the discovery of community structure in real-world benchmark networks.

E. EVALUATION WITH SYNTHETIC BENCHMARK NETWORK

In this section, the efficiency of the introduced algorithm is evaluated in LFR synthetic benchmark networks and is compared with CMA [14], GAOMA-net [13], GACD [11], GATB [12], LabelRank [34], LBLD [29], and CSLPR [17] algorithms. LFR synthetic benchmark networks are presented in [49] and are more suitable with the characteristics of real-world networks. In LFR networks, node degrees distribution and community sizes distribution follow the power law distribution with exponents τ_1 and τ_2 , respectively. The robustness of the community structure is related to the mixing parameter μ . The mixing parameter represents each node's average connections to other communities. The benchmark networks with a smaller mixing parameter have a significant community structure. The values used for different parameters of LFR networks are given below. In the experiments, the network size is set to 500, the maximum and average degree of nodes are set to

50 and 25, respectively, the values of τ_1 and τ_2 are set to 2 and 1, respectively, and the maximum and minimum size of communities are set to 100 and 50, respectively. The mixing parameter μ is considered from 0.00 to 0.50. Considering that the ground-truth community structure is known in LFR synthetic benchmark networks, the NMI criterion is utilized to evaluate the performance of different algorithms. The average NMI obtained from different algorithms is given in Table IX. In the non-deterministic algorithms (CMA, GAOMA-net, GACD, and GATB), each data is the average of 10 runs. The largest NMI value obtained for each network is written in bold. The LCD-SN has obtained the highest NMI value in all networks except for $\mu = \{0.05, 0.10, 0.15\}$. In these networks, LBLD has obtained the highest NMI value. For further comparison, the results of Table IX are given as a plot in Figure 5. As it is observed, by increasing the value of the mixing parameter μ , since the network's

community structure becomes ambiguous, all algorithms' efficiency decreases. This reduction in the LCD-SN, CSLPR, and LBLD is far less than that of other algorithms.

In addition, Table X shows the results of the Friedman test for ranking different algorithms in the LFR synthetic networks. As you can see, the Friedman test statistic value (p-value<0.001) is less than 0.05. Therefore, with a probability of 0.95, it could be mentioned that the average ranks in LFR synthetic networks had statistically significant differences. According to the average ranks, the LCD-SN algorithm won first, and the LBLD and CSLPR algorithms won second and third, respectively. In summary, it can be said that, the LCD-SN algorithm is effective for the discovery of community structure in LFR synthetic benchmark networks.

TABLE VI. Real-world benchmark network information. Thses networks can be downloaded from <https://networkrepository.com>, <https://snap.stanford.edu/data>, <http://konect.cc/networks>, and https://www.cise.ufl.edu/research/sparse/matrices/list_by_id.html websites.

Network	Description	Node	Edge
Karate	Zachary's karate club [36]	34	78
Dolphins	Dolphins social network [33]	62	159
PolBooks	Books about US politics [37]	105	441
Football	American college football network [8]	115	613
SFI	Santa Fe Institute [8]	118	200
NetScience	category of Collaboration Networks [38]	379	914
Email	URV email network [39]	1,133	5,451
PowerGrid	US power grid network [40]	4,941	6,594
PGP	The LCC in the graph of PGP users [41]	10,680	24,316
GrQc	Collaboration network of Arxiv General Relativity [42]	5,242	14,496
ca-AstroPh	Collaboration network of Arxiv Astro Physics [42]	18,772	198,110
ca-HepTh	Collaboration network of Arxiv High Energy Physics Theory [42]	9,877	25,998
ca-HepPh	Arxiv High Energy Physics paper citation network [43]	34,546	421,578
Condmnat-2003	Collaboration network, preprints in condensed matter archive [44]	31,163	240,058
Condmnat-2005	Collaboration network, preprints in condensed matter archive [45]	40,421	351,382
Email Enron	Email communication network from Enron [46]	36,692	183,831
Collaboration	collaboration network, preprints in high-energy physics [47]	8,361	15,751
Internet	structure of internet routers as of July 22, 2006 [48]	22,963	96,872

TABLE VII. Average obtained modularity by different algorithms in real-world benchmark network

Network	CMA	GAOMA-net	GACD	GATB	LabelRank	LBLD	CSLPR	LCD-SN
Karete	0.4188	0.4198	0.4198	0.4127	0.3600	0.3710	0.3715	0.3715
Dolphins	0.5272	0.5267	0.5265	0.5263	0.3735	0.3780	0.4780	0.5005
PolBooks	0.5266	0.5222	0.5265	0.5265	0.4946	0.4560	0.4990	0.5270
Football	0.5908	0.5936	0.5602	0.5836	0.5918	0.5800	0.5860	0.6056
SFI	0.7482	0.7489	0.7503	0.7428	0.7227	0.7200	0.6578	0.6918
NetScience	0.8287	0.8243	0.8021	0.8318	0.8144	0.9400	0.9240	0.8435
Email	0.4326	0.4639	0.3502	0.4606	0.0000	0.5400	0.2990	0.5284
PowerGrid	0.7497	0.7423	0.7073	0.7647	0.5046	0.8200	0.8190	0.9159
PGP	0.6994	0.7146	0.6601	0.7578	0.2394	0.8200	0.8190	0.8629
GrQc	0.7503	0.7105	0.7063	0.7462	0.4361	0.7900	0.7940	0.8277
AstroPh	0.2931	0.2448	0.2529	0.3990	0.0773	0.5000	0.4528	0.5629
ca-HepTh	0.5727	0.5270	0.5348	0.5862	0.1730	0.7000	0.7060	0.7331
ca-HepPh	0.4425	0.4597	0.3823	0.5434	0.1440	0.4900	0.4726	0.6138
Condmnat-2003	0.4898	0.4911	0.4610	0.5696	0.1010	0.7000	0.6640	0.6776
Condmnat-2005	0.4337	0.4877	0.4069	0.4069	0.0763	0.6600	0.6320	0.6311
Email Enron	0.3310	0.2887	0.3030	0.3478	0.0786	0.5500	0.5750	0.5244
Collaboration	0.6838	0.6531	0.6427	0.7036	0.2306	0.7900	0.6251	0.7967

Internet	0.5283	0.5028	0.5075	0.3983	0.0050	0.4900	0.4128	0.4981
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TABLE VIII. The average rank of various algorithms in real-world benchmark networks

Test Statistics		Mean Rank	
N	18	CMA	4.83
Chi-Square	48.751	GAOMA-net	4.47
df	7	GACD	3.53
Asymp. Sig.	0.001	GATB	4.72
		LabelRank	1.56
		LBLD	5.50
		CSLPR	4.81
		LCD-SN	6.58

TABLE IX. NMI value obtained by various algorithms in LFR synthetic benchmark network

μ	CMA	GAOMA-net	GACD	GATB	Label Rank	LBLD	CSLPR	LCD-SN
0.00	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
0.05	0.9988	0.9994	0.9267	1.0000	1.0000	1.0000	0.9846	0.9902
0.10	0.9604	0.9814	0.7335	0.9955	0.9997	1.0000	0.9708	0.9831
0.15	0.8568	0.9811	0.6293	0.9689	0.9576	1.0000	0.9537	0.9652
0.20	0.7271	0.9299	0.5134	0.9317	0.9010	0.9412	0.9382	0.9520
0.25	0.6197	0.7237	0.4207	0.8773	0.7337	0.9127	0.9125	0.9278
0.30	0.4928	0.6333	0.3203	0.8150	0.6493	0.8626	0.8847	0.9044
0.35	0.4084	0.4758	0.2352	0.6014	0.5063	0.8117	0.8132	0.8352
0.40	0.3076	0.3819	0.1966	0.3251	0.4835	0.7159	0.7071	0.7494
0.45	0.2369	0.1977	0.1674	0.2187	0.2000	0.4437	0.5921	0.6147
0.50	0.1717	0.1321	0.1205	0.1528	0.0927	0.4105	0.4414	0.4678

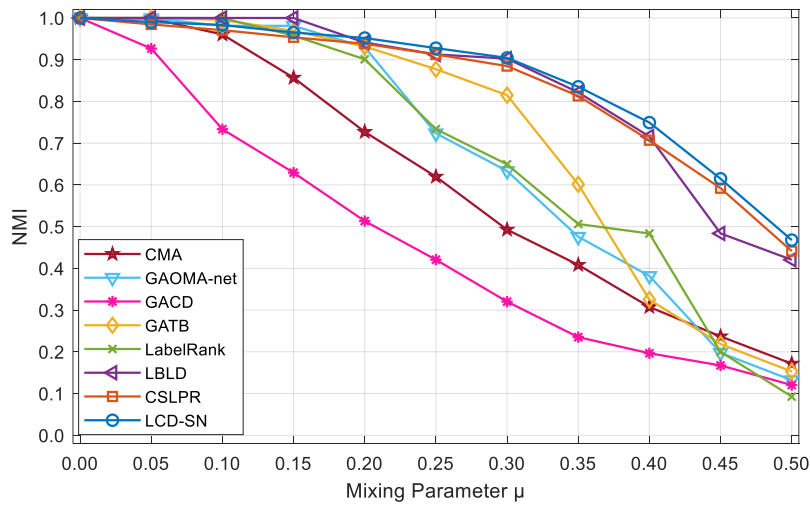


FIGURE 5. NMI value obtained using various algorithms in LFR synthetic benchmark network

TABLE X. The average rank of various algorithms in LFR synthetic benchmark networks

Test Statistics		Mean Rank	
N	11	CMA	2.95
Chi-Square	45.450	GAOMA-net	3.86
df	7	GACD	1.41
Asymp. Sig.	<0.001	GATB	4.95
		LabelRank	4.23
		LBLD	6.59
		CSLPR	5.32
		LCD-SN	6.68

F. RESOLUTION LIMIT

In [50], Lancichinetti et al. have shown that methods based on modularity optimization may fail to determine communities smaller than a specified scale, which is related to the total network size and interconnectedness of the communities. This problem is known as resolution limitation. In this section, the resolution limit of the introduced method is investigated. Consider the network in Figure 6. This network comprises 54 nodes and two communities, C_1 and C_2 , with 50 and 4 nodes, respectively. A node in any community is connected to all other nodes of the same community. In other words, each community is a fully connected graph. The only edge connecting two communities is between node u in community C_1 and node v in community C_2 . The modularity value of the community structure in the network is 0.0097. However, modularity optimization-based approaches assign node u to the community C_2 because this network partition has a larger value of 0.0101 for modularity. This phenomenon is exactly because of the resolution limitation of modularity optimization.

LCD-SN algorithm, along with CMA, GAOMA-net, GATB, GACD, LabelRank, CSLPR, and LBLD algorithms, have been executed 100 times in the test network of Figure 6, and the results are reported in Table XI. Since CMA, GAOMA-net, GACD, and GATB algorithms are based on modularity optimization, in most cases, they wrongly assign node u as a member of the community C_2 . It should be noted that the LabelRank algorithm identifies the entire network as a community in all runs. LCD-SN algorithm has correctly identified communities C_1 and C_2 in all runs. Therefore, our algorithm effectively solves the problem of resolution limitation.

G. DISCUSSION

This subsection explains the reasons behind the introduced algorithm's performance compared to other algorithms. As mentioned, the proposed algorithm first ranks the network nodes. In determining the rank of a node, in addition to its first-degree neighbors, its second-degree neighbors are also considered. Then, starting from the higher-rank nodes, initial local communities consisting of the corresponding node and its first-degree neighbors are formed. This is what happens in community formation in real-world networks. In real-world networks, different nodes usually form a community around important nodes. In short, the introduced algorithm identifies meaningful communities by making full use of the structural information of the network.

V. CONCLUSION AND FUTURE WORKS

This article presents a new method called LCD-SN to detect communities in social networks. The proposed method is a local method based on node ranking. LCD-SN algorithm consists of three phases. In the first phase, network nodes are

ranked using a new criterion called IMP, and initial local communities are formed around high-ranked nodes. In the second phase, overlapping nodes are assigned to a single community using the GLHN similarity measure. In the third phase, the obtained communities in the previous phase are improved by post-processing (removing small communities and merging weak communities with strong ones). Among the advantages of the proposed algorithm its dependence on the minimum number of input parameters, its locality and no need for the information of the entire network, not having the problem of resolution limit, and most importantly, its certainty could be mentioned. In this article, single-layer and unsigned networks are considered. The generalization of the introduced algorithm to identify the community structure in multi-layer, signed and weighted networks could be the focus of our next work.

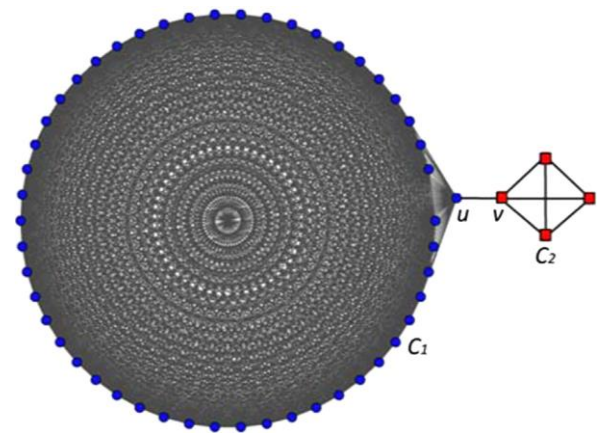


FIGURE 6. Sample network with 54 nodes to investigate the resolution limit

TABLE XI. The resolution limit of different algorithms

Algorithm	The percentage of times that communities C1 and C2 are correctly identified
CMA	53%
GAOMA-net	1%
GACD	22%
GATB	0%
LabelRank	0%
LBLD	46%
CSLPR	31%
LCD-SN	100%

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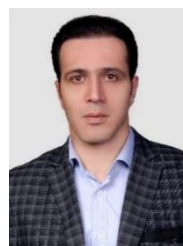
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Jafar Sheykhzadeh was born in Iran in 1977. He received the B.S. degree in computer engineering from the Lahijan branch, Islamic Azad University, Lahijan, Iran, in 2001, and M.S. degree in computer engineering from the Qazvin branch, Islamic Azad University, Qazvin, Iran, in 2009. And now he is a Ph.D. student in computer science at Urmia branch, Islamic Azad University, Urmia, Iran. He is

currently an Instructor in the Department of Computer Engineering, Ahar branch, Islamic Azad University, Ahar, Iran since 2009. His research interests include, task scheduling, complex and social networks analysis.

Bagher Zarei received the B.Sc. degree in computer engineering from the Shabestar branch, Islamic Azad University, Shabestar, Iran, in 2004 and the M.Sc. and Ph.D. degree in computer engineering from the Qazvin branch,



Islamic Azad University, Qazvin, Iran, in 2006 and 2020, respectively. He is currently an Assistant Professor in the Department of Computer Engineering and Information Technology, Shabestar branch, Islamic Azad University, Shabestar, Iran. His research interests include software development, soft computing, complex network analysis, reinforcement learning, and chaos theory.



Farhad Soleimanian Gharehchopogh was born in 1979 in Iran. He received his B.S. in Computer Engineering from Shabestar Branch, Islamic Azad University, Iran, in 2002, his M.S. in Computer Engineering from Cukurova University, Adana/Turkey, in 2011 and his Ph.D. degree in Computer Engineering from Hacettepe University, Ankara/Turkey in 2015. He has been an assistant professor in Computer Engineering at Urmia Branch, Islamic Azad University, Urmia, Iran, since 2015. He is researching metaheuristic algorithms and their application issues. And he published more than 150 papers in international journals and conferences.